UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

77464 7590 12/11/2023

IPUSA, P.L.L.C
330 John Carlyle Street
Suite 220
Alexandria, VA 22314

EXAMINER	
MEGNA FUENTES, ANTHONY W	
ART UNIT	PAPER NUMBER
2855	

DATE MAILED: 12/11/2023

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/653,705	03/07/2022	Takashi KASAI	22MM-004	1791

TITLE OF INVENTION: FLOW SENSOR CHIP

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	$1200	$0.00	$0.00	$1200	03/11/2024

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

77464 7590 12/11/2023

IPUSA, P.L.L.C
330 John Carlyle Street
Suite 220
Alexandria, VA 22314

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/653,705	03/07/2022	Takashi KASAI	22MM-004	1791

TITLE OF INVENTION: FLOW SENSOR CHIP

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	$1200	$0.00	$0.00	$1200	03/11/2024

EXAMINER	ART UNIT	CLASS-SUBCLASS
MEGNA FUENTES, ANTHONY W	2855	073-204240

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

❑ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

❑ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 _____

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ❑ Individual ❑ Corporation or other private group entity ❑ Government

4a. Fees submitted: ❑ Issue Fee ❑ Publication Fee (if required)

4b. Method of Payment: *(Please first reapply any previously paid fee shown above)*

❑ Electronic Payment via the USPTO patent electronic filing system ❑ Enclosed check ❑ Non-electronic payment by credit card (Attach form PTO-2038)

❑ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. **Change in Entity Status** (from status indicated above)

❑ Applicant certifying micro entity status. See 37 CFR 1.29

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

❑ Applicant asserting small entity status. See 37 CFR 1.27

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

❑ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____

UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/653,705	03/07/2022	Takashi KASAI	22MM-004	1791

77464 7590 12/11/2023

IPUSA, P.L.L.C
330 John Carlyle Street
Suite 220
Alexandria, VA 22314

EXAMINER	
MEGNA FUENTES, ANTHONY W	
ART UNIT	PAPER NUMBER
2855	

DATE MAILED: 12/11/2023

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b)
(Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/ owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).
https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf

Routine uses of the information in this record may include disclosure to:

1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
2) a federal, state, local, or international agency, in response to its request;
3) a contractor of the USPTO having need for the information in order to perform a contract;
4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
9) the Office of Personnel Management (OPM) for personnel research purposes; and
10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 17/653,705	Applicant(s) KASAI et al.	
	Examiner ANTHONY W MEGNA FUENTES	Art Unit 2855	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☑ This communication is responsive to the claims filed on 03/07/2022.
   ☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☑ The allowed claim(s) is/are 1,5-6,10-11 and 13-16 . As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see **http://www.uspto.gov/patents/init_events/pph/index.jsp** or send an inquiry to **PPHfeedback@uspto.gov.**

4. ☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

   Certified copies:

   a) ☑ All b) ☐ Some* c) ☐ None of the:

   1. ☑ Certified copies of the priority documents have been received.
   2. ☐ Certified copies of the priority documents have been received in Application No. _____ .
   3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

   * Certified copies not received: _____ .

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
   ☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____ .

   Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☑ Notice of References Cited (PTO-892)
2. ☑ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____.
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____.
4. ☑ Interview Summary (PTO-413), Paper No./Mail Date. _____.
5. ☑ Examiner's Amendment/Comment
6. ☑ Examiner's Statement of Reasons for Allowance
7. ☐ Other _____.

/ANTHONY W MEGNA FUENTES/ Examiner, Art Unit 2855	/HERBERT K ROBERTS/ Primary Examiner, Art Unit 2855

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

Receipt is acknowledged of certified copies of papers required by 37 CFR 1.55.

Election/Restrictions

Applicant's election without traverse of Species 1 in the reply filed on 10/16/2023 is acknowledged.

Claims 3-4, 7 and 12 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected species, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 10/16/2023.

This application is in condition for allowance except for the presence of claims 3-4, 7 and 12 directed to species non-elected without traverse. Accordingly, claims 3-4, 7 and 12 have been cancelled.

Information Disclosure Statement

The information disclosure statements (IDS) were submitted on 03/07/2022 and 09/02/2022. The submission is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

EXAMINER'S AMENDMENT

An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this examiner's amendment was given in an interview with Herman Paris on 11/27/2023.

The application has been amended as follows:

IN THE CLAIMS

Claim 1 (Currently amended): A flow sensor chip comprising: a substrate in which a cavity having an opening is formed; a membrane provided on a surface of the substrate so as to cover the opening; a heater provided inside the membrane; a first thermopile and a second thermopile provided in the membrane, the heater being interposed between the first thermopile and the second thermopile; [[and]] a heat

conduction member serving as a heat conduction path between the substrate and the membrane, wherein a first hot junction and a first cold junction of a thermocouple provided in the first thermopile and a second hot junction and a second cold junction of a thermocouple provided in the second thermopile are arranged at positions overlapping with the opening as viewed from a direction normal to the surface of the substrate; [[, and]] the heat conduction member includes: a first heat conduction member serving as a heat conduction path between the first cold junction and the substrate; and a second heat conduction member serving as a heat conduction path between the second cold junction and the substrate; wherein the first heat conduction member is provided on a surface of the membrane on an opposite side from the substrate, the first heat conduction member extends from an area where the membrane and the substrate overlap with each other to a position closer to the first hot junction than is the first cold junction, as viewed from the direction normal to the surface of the substrate, the second heat conduction member is provided on a surface of the membrane on an opposite side from the substrate, and the second heat conduction member extends from an area where the membrane and the substrate overlap with each other to a position closer to the second hot junction than is the second cold junction, as viewed from the direction normal to the surface of the substrate.

Claim 2 is cancelled.

Claim 3 is cancelled.

Claim 4 is cancelled.

Claim 5 (Currently amended): The flow sensor chip according to claim [[2]] 1, wherein a first through hole that allows communication between an outside and the cavity is formed through the first heat conduction member and the membrane, and a second through hole that allows communication between the outside and the cavity is formed through the second heat conduction member and the membrane.

Claim 7 is cancelled.

Claim 8 is cancelled.

Claim 9 is cancelled.

Claim 12 is cancelled.

Claim 13 (Newly added): A flow sensor chip comprising: a substrate in which a cavity having an opening is formed; a membrane provided on a surface of the substrate so as to cover the opening; a heater provided inside the membrane; a first thermopile and a second thermopile provided in the membrane, the heater being interposed between the first thermopile and the second thermopile; a heat conduction member serving as a heat conduction path between the substrate and the membrane, wherein a first hot junction and a first cold junction of a thermocouple provided in the first thermopile and a second hot junction and a second cold junction of a thermocouple provided in the second thermopile are arranged at positions overlapping with the opening as viewed from a direction normal to the surface of the substrate; the heat conduction member includes: a first heat conduction member serving as a heat conduction path between the first cold junction and the substrate; and a second heat conduction member serving as a heat conduction path between the second cold

junction and the substrate; and wherein the heat conduction member is electrically connected to a ground potential.

Claim 14 (Newly Added): The flow sensor chip according to claim 13, further comprising a metal terminal provided on the substrate, wherein the heat conduction member and the metal terminal are constituted by a same material as each other and have a same film thickness as each other.

Claim 15 (Newly Added): A flow sensor chip comprising: a substrate in which a cavity having an opening is formed; a membrane provided on a surface of the substrate so as to cover the opening; a heater provided inside the membrane; a first thermopile and a second thermopile provided in the membrane, the heater being interposed between the first thermopile and the second thermopile; a heat conduction member serving as a heat conduction path between the substrate and the membrane, wherein a first hot junction and a first cold junction of a thermocouple provided in the first thermopile and a second hot junction and a second cold junction of a thermocouple provided in the second thermopile are arranged at positions overlapping with the opening as viewed from a direction normal to the surface of the substrate; the heat conduction member includes: a first heat conduction member serving as a heat conduction path between the first cold junction and the substrate; and a second heat conduction member serving as a heat conduction path between the second cold junction and the substrate; and wherein the heat conduction member is constituted by a

metal material containing at least one element selected from the group consisting of aluminum, titanium, copper, tungsten, molybdenum, and tantalum.

Claim 16 (Newly Added): The flow sensor chip according to claim 15, further comprising a metal terminal provided on the substrate, wherein the heat conduction member and the metal terminal are constituted by a same material as each other and have a same film thickness as each other.

Reasons for Allowance

Claims 1, 5-6, 10-11 and 13-16 are allowed.

The following is an examiner's statement of reasons for allowance:

Regarding claims 1, 13 and 16, Bonne et al. (***US 6,502,459***; hereinafter "Nishimoto") teaches a flow sensor chip (Figures 1-2) comprising: a substrate (20; Figures 1-2) in which a cavity (30; Figures 1-2) having an opening is formed (the opening of the cavity which is partially covered by members 28-29; Figures 1-2); a membrane (members 28-29; Figures 1-2) provided on a surface of the substrate so as to cover the opening (the members 28-29 are provided on a surface of the substrate 20 to as to cover the opening; Figures 1a-1b); a heater (26; Figures 1-2; Column 6, Lines 39-53) provided inside the membrane (the heater 26 is provided within the members 28-29; Figures 1-2); a first thermopile (22; Figures 1-2) and a second thermopile (24;

Figures 1-2) provided in the membrane (the sensors 22 and 24 are provided in the membrane 28-29; Figures 1-2; Column 6, Lines 39-51), the heater (26; Figures 1-2) being interposed between the first thermopile and the second thermopile (Figures 1-2 demonstrate the heater 26 interposed between the first and second thermopiles 22 and 24); a heat conduction member (openings 82; Figures 1-2) serving as a heat conduction path between the substrate and the membrane (the openings 82 serve as a heat conduction path between the substrate 20 and the membrane 28 and 29; See Figures 1-2), wherein a first hot junction (A; see annotated Figure 2) and a first cold junction (B; see annotated Figure 2) of a thermocouple provided in the first thermopile (22; Figures 1-2) and a second hot junction (C; see annotated Figure 2) and a second cold junction (D; see annotated Figure 2) of a thermocouple provided in the second thermopile (24; Figures 1-2) are arranged at positions overlapping with the opening as viewed from a direction normal to the surface of the substrate (the elements A-D are arranged at positions overlapping with the opening as viewed from a direction normal to the surface of the substrate 20; Figures 1-2); the heat conduction member (openings 82; Figures 1-2) includes: a first heat conduction member (opening 82 adjacent to thermopile 22; Figures 1-2) serving as a heat conduction path between the first cold junction and the substrate (the opening 82 adjacent to thermopile 22 serves as a head conduction path between the first cold junction B and the substrate 20; See Figures 1-2); and a second heat conduction member (opening 82 adjacent to thermopile 24; Figures 1-2) serving as a heat conduction path between the second cold junction and the substrate (the opening 82 adjacent to thermopile 24 serves as a head conduction path between the second cold junction D and the substrate 20; See Figures 1-2).

In claim 1, the specific limitations of "wherein the first heat conduction member is provided on a surface of the membrane on an opposite side from the substrate, the first heat conduction member extends from an area where the membrane and the substrate overlap with each other to a position closer to the first hot junction than is the first cold junction, as viewed from the direction normal to the surface of the substrate, the second heat conduction member is provided on a surface of the membrane on an opposite side from the substrate, and the second heat conduction member extends from an area where the membrane and the substrate overlap with each other to a position closer to the second hot junction than is the second cold junction, as viewed from the direction normal to the surface of the substrate" in combination with the remaining limitations as claimed are neither anticipated nor made obvious over the prior art made of record.

Claims 5-6 and 10-11 are also allowed for depending on claim 1.

In claim 13, the specific limitations of "wherein the heat conduction member is electrically connected to a ground potential" in combination with the remaining limitations as claimed are neither anticipated nor made obvious over the prior art made of record.

Claim 14 would also allowed for depending on claim 13.

In claim 15, the specific limitations of "wherein the heat conduction member is constituted by a metal material containing at least one element selected from the group consisting of aluminum, titanium, copper, tungsten, molybdenum, and tantalum" in

combination with the remaining limitations as claimed are neither anticipated nor made obvious over the prior art made of record.

Claim 16 would also allowed for depending on claim 15.

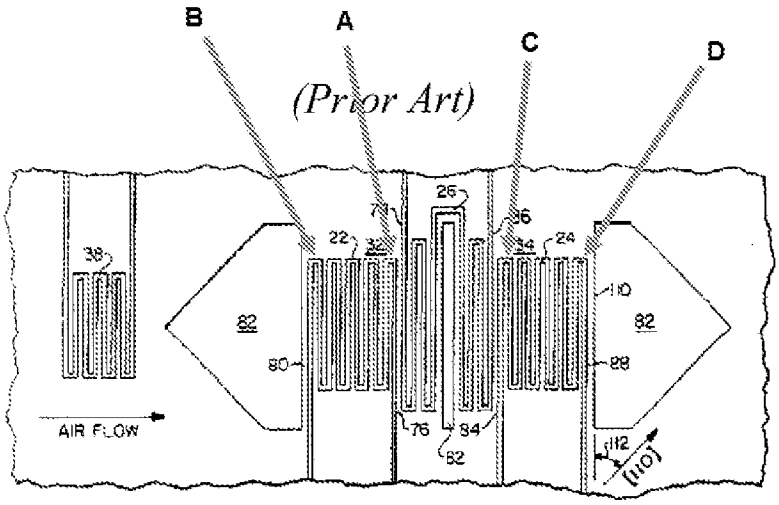


FIG. 2

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

- Speldrich (US 8,286,478)

- Nagata et al. (US 5,291,781)

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ANTHONY W MEGNA FUENTES whose telephone number is (571)272-6456. The examiner can normally be reached M-F: 8AM-4PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Peter Macchiarolo can be reached on 571-272-2375. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197

(toll-free). If you would like assistance from a USPTO Customer Service

Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ANTHONY W MEGNA FUENTES/
Examiner, Art Unit 2855

/HERBERT K ROBERTS/
Primary Examiner, Art Unit 2855